

DWIJ PRAJAPATI

Aspiring AI Engineer

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PROFESSIONAL SUMMARY

AI, Backend & Full-Stack Engineer with extensive experience designing, building, and deploying end-to-end Generative AI applications, intelligent systems, and scalable backend platforms. Proven expertise in building low-latency asynchronous microservices, robust API architectures, and responsive web applications. Skilled across the full lifecycle—from architecting high-performance Python backends (FastAPI, Django, Asyncio) and distributed state management (Redis) to multi-step LLM orchestration, agentic workflows, and automated evaluation engines.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, JavaScript, SQL, Bash/Shell

AI / Machine Learning: LLMs, Agentic Architectures, Prompt Engineering & Scaffolding, LLM Evaluation & Scoring Pipelines, Multi-Turn Conversation Systems, Speech AI (STT/TTS & VAD), Scikit-learn, TensorFlow, NLP

Backend & System Design: FastAPI, Django, Asyncio, RESTful API Design, WebSockets, Webhook Architecture, Distributed Session Management, Pydantic, Event-Driven Architecture, HTTPX

Databases & State: Redis (In-Memory Caching & Session Checkpointing), MySQL, PostgreSQL

Cloud, DevOps & Tools: Docker, Docker Compose, AWS (EC2, S3), Caddy (Reverse Proxy/TLS), Git, GitHub, Pytest (E2E & Unit Testing), Postman

PROFESSIONAL EXPERIENCE

AI Developer Intern, Banao Technologies (InterviewGod.ai)

Feb 2026 – Present

Projects: InterviewGod.ai (Real-Time Voice AI Agent & LLM Evaluation Engine) and Multi-Modal Candidate Screening & Decision Governance Pipeline

- Architected an **asynchronous event-driven** real-time voice AI pipeline leveraging **LiveKit SFU (WebRTC)**, **OpenAI GPT-4o**, **ElevenLabs**, and **Sarvam AI**, implementing semantic turn-taking, acoustic VAD filtering, and pre-generated audio warmup gates to achieve **sub-second conversational latency**.
- Engineered an **adaptive multi-turn dialog** state machine with heuristic repeat-request and filler-word interceptors, enforcing bounded per-topic counter-probing and contextual conversational scaffolding to eliminate topic drift and prevent task stalls.
- Developed an **LLM-as-a-Judge answer-validity gate** (gpt-4o-mini) with fail-open resiliency to intercept STT hallucinations and non-answers, resolving mathematical denominator inflation bugs in weighted skill aggregation algorithms by penalizing invalid responses and normalizing zero-score distributions.
- Engineered a **distributed session state** and checkpointing architecture using **Redis** with 24-hour handoff buffers, enabling **zero-data-loss** disconnect/reconnect recovery for active sessions alongside bidirectional webhook synchronization to core microservices.
- Audited a **four-layer deterministic candidate evaluation** and call-screening pipeline (Scenario Detection, Telemetry, Semantic Intelligence, Policy Engine, Governance), resolving **20+ edge cases across multilingual** (Hindi/English) speech pipelines, brittle speaker-attribution logic, telemetry constant drifts, and silent fallbacks.
- Constructed a **multi-modal weighted aggregation** engine consolidating Resume, Screening Call, Assessment, Technical Interview, and Integrity signals into automated **PROCEED / HOLD / REJECT** hiring recommendations with calibrated confidence scoring.
- Containerized and deployed multi-service production stacks across **AWS EC2 utilizing Docker Compose and Caddy reverse** proxies with automated **SSL/TLS termination**, supporting multi-stage Serverless framework migrations across AWS accounts with IAM OIDC role delegations and CodeBuild CI/CD pipelines.

PROJECTS

FinSight AI — Quantitative Intelligence & Autonomous Trading Platform — [Link](#)

- Engineered time-series models combining **SARIMAX (ADF stationarity testing & seasonal decomposition)** and regularized **LSTM/MLP neural networks with FinBERT NLP sentiment analysis** and cross-asset macroeconomic indicators (S&P 500, VIX, 10-Yr Yields) to capture non-linear market dynamics and momentum shifts.
- Built an autonomous Markov Decision Process (MDP) stock trading environment supporting discrete and continuous action spaces; trained **PPO, A2C, and DQN agents** on dynamic multi-factor state spaces (14-day RSI, MACD, Bollinger Bands) to optimize risk-adjusted trading strategies.
- Implemented Markowitz Modern Portfolio Theory (MPT) using **SLSQP numerical optimization and Monte Carlo simulations** to generate the Efficient Frontier (maximizing Sharpe Ratio), while modeling downside exposure **Historical/Parametric VaR, CVaR (Expected Shortfall), and macroeconomic stress testing**.

CompInsight AI — Enterprise ML Compensation Modeling & Valuation Engine — [Link](#)

- Engineered a production-grade ML pipeline benchmarking Gradient Boosted Decision Trees (**XGBoost, LightGBM, CatBoost**) alongside Ridge/Random Forest regressors; implemented log-normal target transformations for heavy-tailed compensation distributions, target/one-hot encoding for high-cardinality features, and **5-Fold Cross-Validation with Bayesian hyperparameter optimization**, achieving an **R² score of 0.93** and a **22% reduction in RMSE**.
- Integrated **SHAP (SHapley Additive exPlanations)** utilizing TreeExplainer and Partial Dependence Plots (PDP) to quantify non-linear feature interactions and marginal contributions (e.g., tech stack scarcity, tenure elasticity, geographic tiering), ensuring algorithmic transparency and bias mitigation in salary benchmarking.
- Built a decoupled, container-ready backend using **FastAPI with Pydantic** data contracts, input sanitation, and serialized preprocessing/model DAGs (Joblib); achieved **sub-40ms P99 inference** latency across RESTful endpoints with automated OpenAPI schemas integrated into a reactive Streamlit executive dashboard.

EDUCATION

B.Tech in Computer Engineering, G.H. Patel College of Engineering, Anand

2022 – 2026

CGPA: 8.18 / 10

ACHIEVEMENTS

- Machine Learning A-Z: AI, Python & R + ChatGPT – Prize
- Complete Python Bootcamp – Certification